Cambridge Assessment International Education

# Cambridge International AS & A Level

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*5268134660*

**MATHEMATICS** **9709/11**

Paper 1 Pure Mathematics 1 **October/November 2023**

**1 hour 50 minutes**

You must answer on the question paper.

You will need: List of formulae (MF19)

**INSTRUCTIONS**

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.

**INFORMATION**

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **20** pages. Any blank pages are indicated.


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**1** **(a)** Expand $(1 + 3x)^6$ in ascending powers of $x$ up to, and including, the term in $x^2$. [2]

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**(b)** Hence find the coefficient of $x^2$ in the expansion of $(1 - 7x + x^2)(1 + 3x)^6$. [2]

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**2** A line has equation $y = 2cx + 3$ and a curve has equation $y = cx^2 + 3x - c$, where $c$ is a constant.

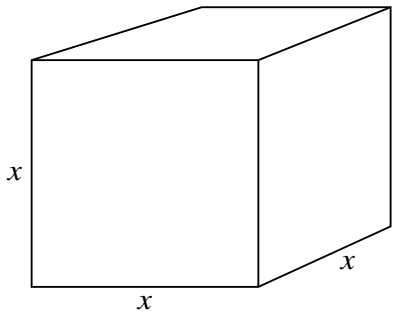
Showing all necessary working, determine which of the following statements is correct.

**A** The line and curve intersect only for a particular set of values of $c$.

**B** The line and curve intersect for all values of $c$.

**C** The line and curve do not intersect for any values of $c$. [4]

**3**




The diagram shows a cubical closed container made of a thin elastic material which is filled with water and frozen. During the freezing process the length, $x$ cm, of each edge of the container increases at the constant rate of 0.01 cm per minute. The volume of the container at time $t$ minutes is $V$ cm$^3$.

Find the rate of increase of $V$ when $x = 20$. [3]

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**4** The transformation R denotes a reflection in the $x$-axis and the transformation T denotes a translation of $\begin{pmatrix} 3 \\ -1 \end{pmatrix}$.

**(a)** Find the equation, $y = g(x)$, of the curve with equation $y = x^2$ after it has been transformed by the sequence of transformations R followed by T. [2]

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**(b)** Find the equation, $y = h(x)$, of the curve with equation $y = x^2$ after it has been transformed by the sequence of transformations T followed by R. [2]

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**(c)** State fully the transformation that maps the curve $y = g(x)$ onto the curve $y = h(x)$. [2]

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**5** **(a)** Show that the equation

$$4\sin x + \frac{5}{\tan x} + \frac{2}{\sin x} = 0$$

may be expressed in the form $a\cos^2 x + b\cos x + c = 0$, where $a$, $b$ and $c$ are integers to be found. [3]

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**(b)** Hence solve the equation $4\sin x + \frac{5}{\tan x} + \frac{2}{\sin x} = 0$ for $0° \leqslant x \leqslant 360°$. [3]

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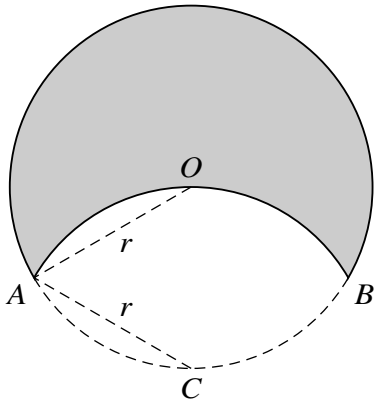
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**6**




The diagram shows a motif formed by the major arc $AB$ of a circle with radius $r$ and centre $O$, and the minor arc $AOB$ of a circle, also with radius $r$ but with centre $C$. The point $C$ lies on the circle with centre $O$.

**(a)** Given that angle $ACB = k\pi$ radians, state the value of the fraction $k$. [1]

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**(b)** State the perimeter of the shaded motif in terms of $\pi$ and $r$. [1]

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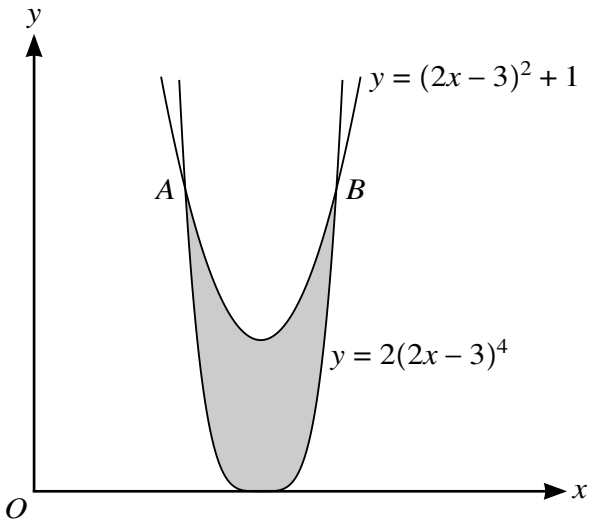
**(c)** Find the area of the shaded motif, giving your answer in terms of $\pi$, $r$ and $\sqrt{3}$. [5]

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**7** The sum of the first two terms of a geometric progression is 15 and the sum to infinity is $\frac{125}{7}$. The common ratio of the progression is negative.

Find the third term of the progression. [7]

**8**




The diagram shows the curves with equations $y = 2(2x - 3)^4$ and $y = (2x - 3)^2 + 1$ meeting at points $A$ and $B$.

**(a)** By using the substitution $u = 2x - 3$ find, by calculation, the coordinates of $A$ and $B$. [4]

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**(b)** Find the exact area of the shaded region. [5]

**9** **(a)** Express $4x^2 - 12x + 13$ in the form $(2x + a)^2 + b$, where $a$ and $b$ are constants. [2]

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The function f is defined by $f(x) = 4x^2 - 12x + 13$ for $p < x < q$, where $p$ and $q$ are constants. The function g is defined by $g(x) = 3x + 1$ for $x < 8$.

**(b)** Given that it is possible to form the composite function gf, find the least possible value of $p$ and the greatest possible value of $q$. [3]

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**(c)** Find an expression for $\mathrm{gf}(x)$. [1]

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The function h is defined by $\mathrm{h}(x) = 4x^2 - 12x + 13$ for $x < 0$.

**(d)** Find an expression for $\mathrm{h}^{-1}(x)$. [3]

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**10** A curve has a stationary point at $(2, -10)$ and is such that $\frac{d^2y}{dx^2} = 6x$.

**(a)** Find $\frac{dy}{dx}$. [3]

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**(b)** Find the equation of the curve. [3]

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**(c)** Find the coordinates of the other stationary point and determine its nature. [3]

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**(d)** Find the equation of the tangent to the curve at the point where the curve crosses the $y$-axis. [2]

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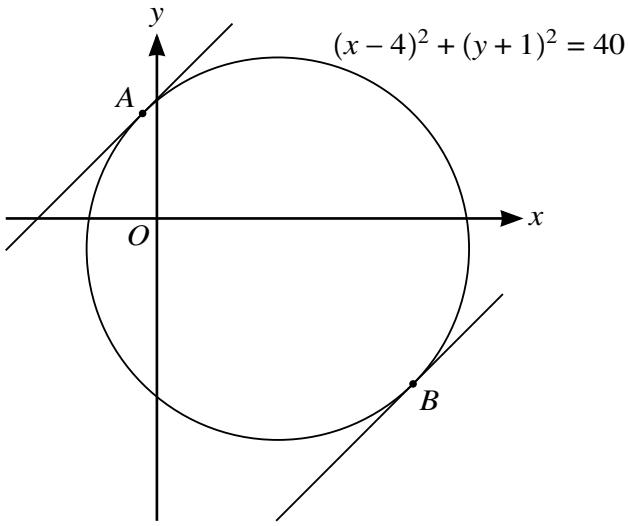
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**11**




The diagram shows the circle with equation $(x-4)^2+(y+1)^2=40$. Parallel tangents, each with gradient 1, touch the circle at points $A$ and $B$.

**(a)** Find the equation of the line $AB$, giving the answer in the form $y=mx+c$. [3]

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**(b)** Find the coordinates of $A$, giving each coordinate in surd form. [4]

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**(c)** Find the equation of the tangent at $A$, giving the answer in the form $y = mx + c$, where $c$ is in surd form. [2]

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## Additional Page

If you use the following lined page to complete the answer(s) to any question(s), the question number(s) must be clearly shown.

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